DRAINAGE

Highway drainage facilities carry water across the right-of-way and remove stormwater from the roadway itself. Drainage facilities include;

- bridges,
- culverts,
- channels,
- curbs,
- gutters, and
- various types of drains.

Hydraulic capacities and locations of such structures should be designed to take into consideration;

- damage to upstream and downstream property and
- to secure as low a degree of risk of traffic interruption by flooding as is consistent with the importance of the road, the design traffic service needs.

Hydrologic Analysis of Drainage for Highway

Whenever there is a rainfall, some of the rain water infiltrated into the ground and stored as ground water and some of the portion may evaporate into the atmosphere. Other than these losses, the water left on the surface is called as **run off**.

The method of estimating the run off is called **hydrologic analysis**. To estimate the maximum quantity of water expected to reach the drainage system is the main objective of hydrologic analysis.

The factors affecting run off and they are

- Rate of rain fall
- Moisture condition
- Soil type
- Ground cover presence
- Topography

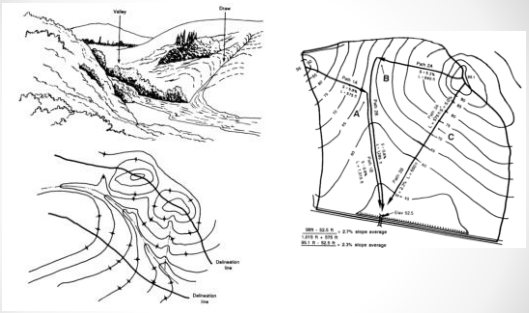
Other than the above factors, rain fall intensity, occurrence of storms in that area are to be studied from the old records. Hence, maximum run off can be estimated to build safe surface drainage system.

The run off can be calculated by below formula

$$\mathbf{Q = C * i * } A_d$$

Q = run off (m^3/sec)
C = run off coefficient
i = intensity of rain fall (mm/sec)
A_d = area of drainage (m^2)

Area of drainage



The run off can be calculated by below formula

$$\mathbf{Q = C * i * A_d}$$

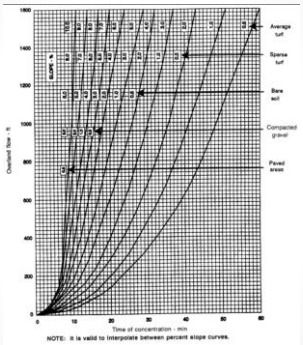
Q = run off (m^3/sec)
C = run off coefficient
i = intensity of rain fall (mm/sec)
A_d = area of drainage (m^2)

Type of Surface	Coefficient of run off
Pervious soil surface	0.05 – 0.30
Soil covered with turf	0.30 – 0.55
Impervious soil	0.40 – 0.65
Gravel & WBM roads	0.35 – 0.70
Bituminous & C.C roads	0.80 – 0.90

If the drainage area contains different surfaces in it then run off coefficient is calculated as;

$$\mathbf{C} = \frac{\boldsymbol{A_1 * C_1 + A_2 * C_2 + A_3 * C_3}}{\boldsymbol{A_1 + A_2 + A_3}}$$

Where C_1, C2, C3 are run off coefficients for different surfaces and A_1, A2, A3 are their respective areas.



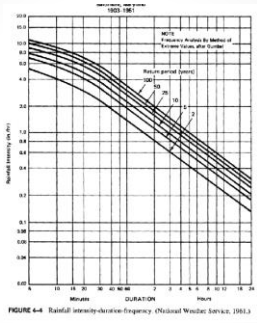


FIGURE 4-4 Rainfall intensity-duration-frequency. (National Weather Service, 1961.)

$$Q = V * A$$

Q = run off (m^3/sec)
V = Allowable velocity in channel (m/sec)
A = cross section area of channel

Soil type	Allowable velocity (m/sec)
Sand or silt	0.30 – 0.50
Loam	0.60 – 0.90
Clay	0.90 – 1.50
Gravel	1.20 – 1.50
Soil with grass	1.50 – 1.80



Allowable Velocity

Manning's formula:

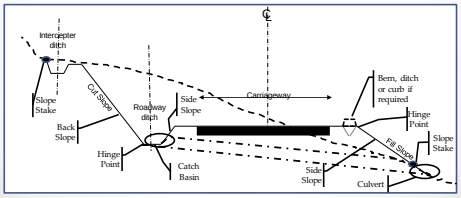
$$V = \frac{1}{n} * R^{\frac{2}{3}} * S^{\frac{1}{2}}$$

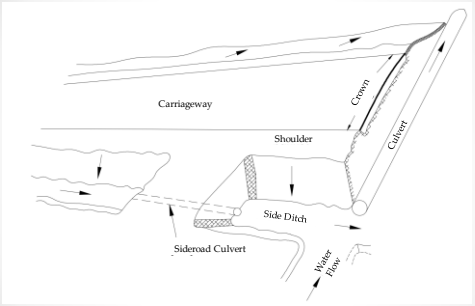
V = Allowable velocity in channel (m/sec)
n = Manning's roughness coefficient
R = Hydraulic radius (m) :the ratio of area of the channel to its wetted perimeter
S = Longitudinal Slope of Channel

Lining material	Manning's roughness coefficient, n
Ordinary soil	0.02
Soil with grass layer	0.05 – 0.10
Concrete lining	0.013
Rubble lining	0.04

DRAINAGE

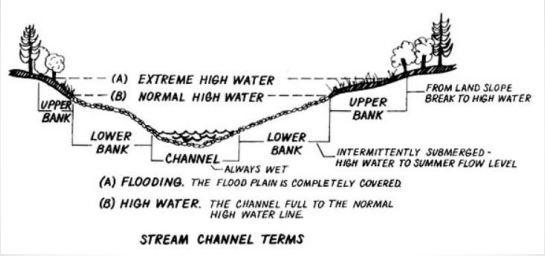
Ditch: A small artificial channel cut through earth or rock to carry water for irrigation or drainage. A long narrow cut made in the earth to bury pipeline, cable, or similar installations.
Slope, crossfall: Surface cross slopes are necessary on all components of the crosssection to facilitate drainage.
Slope stake: Slope staking is a trial and error process of finding the point where the slope intercepts the natural ground. This point is referred to as the "catch point" or "slope catch."
Intercepting ditch: An interceptor ditch is a small ditch or channel constructed to intercept and convey water to an area where it can be safely discharged.
Side slope: A long narrow hole that is dug along a road, field, etc., and used to hold or move water; a long narrow hole that is dug along a road, field, etc., and used to hold or move water.



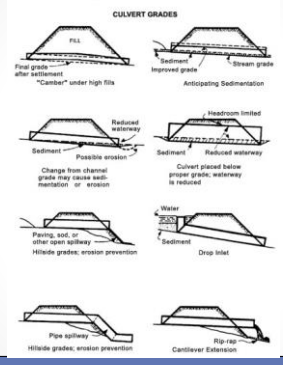


11

DRAINAGE



DRAINAGE



DRAINAGE

